SRS — Belle Biens Digital Platform (Web, Mobile App, Admin Dashboard)

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Document control

This SRS covers one platform for Belle Biens (@bellebiens), a floral café and bakery in Damascus: a customer website, a Flutter mobile app, and an admin dashboard on one shared backend. Requirements are written in English; explanations are in Arabic. القرارات التي اتخذتها باستقلالية مسجّلة في الملحق D والنقاط المفتوحة في الملحق E.

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Date	2026-11-10

Revision history

Version	Date	Author	Description of change
0.1	2026-11-10	Mohammed Saleh Samman	First full draft from the analysis conversation, with decisions on platform, designer scope, payments and language

Approvals

Name	Role	Signature	Date
[Name]	Client representative, Belle Biens		
[Name]	Project manager		
[Name]	Technical lead		

1. Introduction

The platform turns Belle Biens' Instagram audience into online orders, with an interactive bouquet designer as the main differentiator.

1.1 Purpose

This document specifies the requirements for the Belle Biens Digital Platform, version 1.0. It is written for the client (shop owner and staff), the developers, the UI/UX designer, the testers and the project supervisor. هي المرجع الوحيد لما يجب بناؤه واختباره وتسليمه.

1.2 Scope

Product name: Belle Biens Digital Platform.

The system shall provide:

- A customer website (Progressive Web App) and a customer mobile app (Flutter) to browse and order flowers, cakes and pastries, and gifts.
- A bouquet designer in three stages: layered 2D preview from cut-out photos, interactive 3D preview, and an AI-generated realistic preview.
- Cakes and sweets customization, and a split box builder for flower-and-sweets boxes.
- Cart, checkout, scheduled delivery inside Damascus, gift card messages, and order tracking.
- Payments by cash on delivery, Sham Cash, and Stripe (for customers abroad).
- WhatsApp hand-off and notifications, and local SEO for the website.
- An admin dashboard for products, stock, orders, prices, coupons, delivery zones, content and reports.

The system shall not provide (out of scope for v1.0):

- In-store point of sale, accounting or payroll.
- Supplier and purchasing management.
- A dedicated delivery-driver app. Drivers are handled through order status updates in the dashboard.
- Loyalty points and subscriptions (listed as future work).
- Delivery outside Damascus and multi-branch operation. The data model keeps a branch field so these can be added later.
- Integration with the Movo and BeeOrder platforms beyond outbound links.

Business objectives: convert Instagram traffic into orders on mobile; let customers design a bouquet themselves before ordering; reduce back-and-forth on WhatsApp by sending the shop a complete "bouquet recipe" with each order. أهداف رقمية متفق عليها مع العميل تُسجّل في الملحق E (OI-01).

1.3 Definitions, acronyms and abbreviations

Term	Definition
SRS	Software Requirements Specification
FR / NFR	Functional / Non-functional requirement

Term	Definition
PWA	Progressive Web App: a website installable on a phone and usable on slow networks
RTL	Right-to-left layout used for Arabic
Bouquet designer	The tool where the customer composes a bouquet and sees a live preview
Smart Slot	A predefined position (x, y, scale, rotation, layer) where a flower image is placed in the 2D bouquet preview
Bouquet spec	A JSON document that fully describes a designed bouquet (see Appendix B)
Bouquet recipe	A printable work sheet for the florist generated from a bouquet spec
Split box	A gift box divided into compartments, each filled with flowers or sweets
Daily flowers	The list of fresh flowers available today, with limited stock, maintained by staff
Cut-out image	A photo of one flower, wrap or extra on a transparent background (PNG or WebP)
GLB / glTF	A compressed 3D model format used for the 3D preview
AI preview	A generated realistic image of the designed bouquet
Sham Cash	A Syrian mobile payment service used as an online payment method
COD	Cash on delivery
MoSCoW	Priority scale: Must, Should, Could, Won't (this release)
OTP	One-time password sent to a phone number

1.4 References

ID	Document	Version / date
R1	ISO/IEC/IEEE 29148, Requirements engineering	2018
R2	ISO/IEC 25010, Product quality model	2011 / 2023
R3	Analysis conversation: Belle Biens website and bouquet designer	2026-11-10
R4	Instagram page @bellebiens (public information, working hours, branch, delivery platforms)	Reviewed in R3

ID	Document	Version / date
R5	WCAG, Web Content Accessibility Guidelines	2.1
R6	OWASP Top 10 and ASVS	Current at implementation

1.5 Document overview

Section 2 gives the context, user classes, constraints and release plan. Section 3 lists interfaces, functional requirements and use cases. Section 4 lists quality requirements. Section 5 covers data, constraints, legal points and the photo-shoot content needed. Section 6 defines verification. The appendices hold traceability, the bouquet JSON schema, risks, decisions and open issues.

2. Overall description

This section explains the context behind the requirements in Section 3 and contains no testable requirements itself.

2.1 Product perspective

Belle Biens is a floral café, bakery and gift shop in Abu Rummaneh, Damascus, open daily from 10:00 to 24:00 according to its Instagram page. Today, customers discover products on Instagram (about 15,000 followers) and order through delivery platforms (Movo, BeeOrder) or WhatsApp. The new platform is a new, standalone system: it does not replace an existing system, and it links to Instagram, WhatsApp and the delivery platforms.

Three clients share one backend: the customer website (PWA), the customer mobile app (Flutter), and the admin dashboard (web). The architecture is drawn in 2.8.

2.2 Product functions (summary)

- Browse flowers, cakes and pastries, gifts and ready-made bouquets; filter by occasion, color, flower type, price and availability.
- "Daily flowers": today's fresh stock, updated by staff.
- Bouquet designer with 2D, 3D and AI preview, producing a bouquet spec and a florist recipe.
- Cake and sweets customization; split box builder for flower-and-sweets boxes.
- Cart, checkout as guest or account holder, delivery scheduling, gift card.
- Payments: cash on delivery, Sham Cash, Stripe for customers abroad.
- Order tracking, WhatsApp confirmation and notifications.
- Admin dashboard for catalog, stock, orders, prices, coupons, zones, content and reports.

2.3 User classes and characteristics

User class	Description	Technical skill	Frequency of use	Priority
Customer in Damascus	Orders flowers or cakes for a gift or event, mostly from a phone,	Low	Occasional, peaks at Valentine's	Primary

User class	Description	Technical skill	Frequency of use	Priority
	often arriving from Instagram		Day, Mother's Day, graduation	
Customer abroad	Expatriate sending a gift to someone in Damascus, pays by card	Medium	Occasional	Primary
Shop owner / manager	Sets prices, products, coupons and zones; reads reports	Medium	Daily	Primary
Florist / kitchen staff	Updates daily stock, prepares orders from recipes	Low	Daily, many times	Primary
Delivery coordinator	Assigns and updates deliveries	Low	Daily	Secondary
Developer / support	Operates the system	High	As needed	Secondary

2.4 Operating environment

- Website: current and previous major versions of Chrome, Safari, Edge and Firefox; mobile browsers on Android 9+ and iOS 15+; designed mobile-first.
- Mobile app: Flutter app for Android 9+ (API 28) and iOS 15+.
- Dashboard: desktop and tablet browsers, used in the shop.
- Network: unstable mobile connections are common in Syria, so pages and images must stay light.
- Hosting: cloud hosting with a CDN. The provider is open item OI-04.

2.5 Constraints

- The shop delivers only inside Damascus in v1.0.
- Online payment options are limited by what is legally and technically available in Syria (see Appendix C, risk R-01).
- Real photos of flowers, wraps and extras must be produced by the shop (see 5.6); the designer cannot launch without them.
- Arabic with full RTL is mandatory; English is also required.

2.6 Assumptions and dependencies

ID	Type	Statement	Impact if false
A-01	Assumption	Most customers will arrive from	Mobile performance targets lose

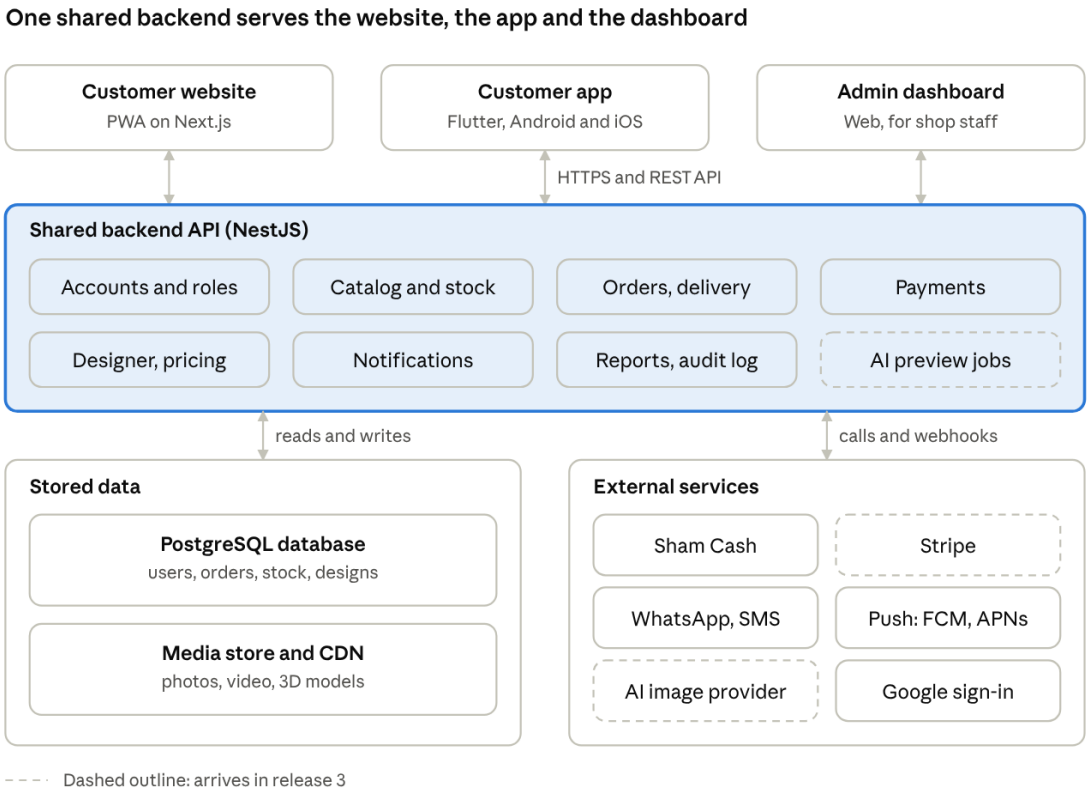
ID	Type	Statement	Impact if false
		Instagram on a phone	their priority
A-02	Assumption	Staff can update daily flower stock once or twice a day	"Daily flowers" and stock-based availability become unreliable
A-03	Assumption	The shop will deliver the photo-shoot content listed in 5.6 before the designer is built	The designer is delayed or must use placeholder art
D-01	Dependency	Sham Cash offers a way to confirm payments (API, callback or manual check)	Fallback manual verification applies (FR-PAY-03)
D-02	Dependency	A legal way exists to accept Stripe payments for a Syrian business	Stripe is dropped or moved to a partner entity (R-01)
D-03	Dependency	A WhatsApp Business account is available for the shop	Notifications fall back to wa.me links and SMS
D-04	Dependency	An image-generation provider is available and affordable	AI preview is postponed without affecting 2D and 3D

2.7 Release plan

All three designer stages are in scope of this SRS (client decision). They are delivered in releases so the shop can start selling early. Durations are estimated after the UI/UX design is approved (OI-02).

Release	Scope	Priority
R1: Web MVP	Backend and database, admin dashboard, website store, cart and checkout, COD and Sham Cash, delivery zones, WhatsApp hand-off, 2D bouquet designer, cakes and sweets, split box builder, SEO, analytics	Must
R2: Mobile app	Flutter app with catalog, 2D designer, cart, orders, push notifications, saved addresses and occasion reminders	Must
R3: Advanced designer and payments	3D preview on web and app, AI realistic preview, Stripe payments, reviews and Instagram content blocks	Should

2.8 System context and architecture



architecture · 3 clients, 1 backend, data and outside services

The three clients talk only to the shared backend. The backend owns the price, stock and order rules, and is the only part that touches the database, the media store and outside services, so every client sees the same results.

3. Specific requirements

Each requirement has a unique ID, a MoSCoW priority, a release (R1, R2, R3 from 2.7) and a verification method (T = Test, I = Inspection, A = Analysis, D = Demonstration). أغلب المتطلبات مصدرها المحادثة (R3)، وأضفتُ ما يلزم لإكمال النظام (الحسابات، الأمان، المخزون، التقارير، الامتثال)؛ وهذه الإضافات مسجّلة في الملحق D.

3.1 External interface requirements

ID	Interface	Requirement	Priority	Rel.	Ver.
EI-UI-01	User interface	The website and app shall display all customer screens in Arabic (RTL) and English (LTR), switchable at any time without losing the cart or the bouquet in progress.	Must	R1	T

ID	Interface	Requirement	Priority	Rel.	Ver.
EI-UI-02	User interface	Customer screens shall follow a mobile-first layout: two-column product grid, horizontal category strip, product card with image, name, price and add button, and a fixed bottom cart button.	Must	R1	I
EI-UI-03	User interface	The bouquet designer screen shall show the live preview and price at the top (or side on wide screens), the component options below (or beside), and a fixed "Add to cart" button.	Must	R1	I
EI-UI-04	User interface	Every customer page shall show visible "Order now", WhatsApp and map-location actions, or reach them in one tap.	Must	R1	I
EI-UI-05	User interface	The visual identity shall follow Belle Biens: soft pink, cream, deep red and soft green, with an elegant Arabic and Latin typeface pair.	Should	R1	I
EI-UI-06	User interface	The admin dashboard shall be usable on a 10-inch tablet and on desktop.	Must	R1	D
EI-HW-01	Hardware	The dashboard shall print the bouquet recipe and the order slip on A4 and on an 80 mm thermal printer through the browser print function.	Should	R1	D
EI-HW-02	Hardware	The mobile app shall use the phone camera only for an optional payment receipt photo.	Could	R2	D
EI-SW-01	Software	The system shall hand an order or a bouquet spec to WhatsApp through a pre-filled wa.me message to the shop number.	Must	R1	T
EI-SW-02	Software	The system shall send order notifications through the WhatsApp Business Platform, with SMS as fallback. Provider: OI-05.	Should	R1	T
EI-SW-03	Software	The system shall integrate Sham Cash for online payment through a payment-provider interface (see FR-PAY-03).	Must	R1	T
EI-SW-04	Software	The system shall integrate Stripe Checkout	Should	R3	T

ID	Interface	Requirement	Priority	Rel.	Ver.
		for card payments in USD.			
EI-SW-05	Software	The system shall call an image-generation service from the server only, through a provider-independent interface.	Should	R3	T
EI-SW-06	Software	The system shall support sign-in with Google and with phone OTP, and shall deliver OTP codes through an SMS or WhatsApp provider (OI-05).	Should	R1	T
EI-SW-07	Software	The system shall show the branch on an embedded map and open the native maps app for directions.	Must	R1	D
EI-SW-08	Software	The website shall embed Instagram posts and Reels, and shall link to @bellebiens.	Could	R3	D
EI-SW-09	Software	The mobile app shall receive push notifications through FCM (Android) and APNs (iOS).	Must	R2	T
EI-SW-10	Software	The system shall serve images, video and 3D models through a CDN-backed media store (Cloudinary or S3-compatible).	Must	R1	I
EI-SW-11	Software	The system shall send analytics events to GA4 and Meta Pixel on the website, and to Firebase Analytics in the app, only after the user's consent.	Should	R1	T
EI-SW-12	Software	The dashboard shall export orders and reports as CSV.	Should	R1	T
EI-CM-01	Communicatio ns	All client-server traffic shall use HTTPS with TLS 1.2 or higher.	Must	R1	I
EI-CM-02	Communicatio ns	The backend shall expose a versioned REST API described by an OpenAPI document and used by all three clients.	Must	R1	I
EI-CM-03	Communicatio ns	Payment and messaging providers shall call the backend through signed webhooks that the backend verifies.	Must	R1	T

3.2 Functional requirements

3.2.1 Catalog, home page and product pages

ID	Requirement	Priority	Rel.	Ver.
FR-CAT-01	The system shall organize products into the categories Fresh flowers, Ready-made bouquets, Cakes and pastries, Sweets boxes, Gifts, and Coffee and matcha add-ons.	Must	R1	I
FR-CAT-02	The system shall let a customer filter flowers and bouquets by occasion, color, flower type, price range, delivery availability and seasonal availability.	Must	R1	T
FR-CAT-03	The system shall let a customer filter cakes and sweets by size, flavor, number of servings and price range.	Must	R1	T
FR-CAT-04	The system shall search products in Arabic and English, ignoring diacritics and treating letter variants (أ إ ا, ة ه, ى ي) as equal.	Should	R1	T
FR-CAT-05	The product page shall show several images, a clear price, the available options, stock status, the delivery policy and a WhatsApp button.	Must	R1	I
FR-CAT-06	The product page may show a short video of the product.	Could	R3	D
FR-CAT-07	The system shall let a customer open any ready-made bouquet in the bouquet designer as a starting point ("Customize this bouquet").	Should	R1	D
FR-CAT-08	The system shall suggest add-ons (gifts, chocolates, coffee, matcha) on the product page and in the cart.	Should	R1	D
FR-CAT-09	The home page shall present a hero with video or images, Today's flowers, featured sweets, gifts, customer reviews, branch details and working hours.	Must	R1	I
FR-CAT-10	The home page and the main menu shall show a prominent "Design your bouquet" action.	Must	R1	I
FR-CAT-11	The system shall offer occasion pages (birthday, engagement, graduation, newborn, anniversary, thank you) listing matching ready-made bouquets and gifts.	Should	R1	D

3.2.2 Daily flowers

قسم "ورد اليوم" يتغير يوميًا حسب الورد الطازج المتوفر.

ID	Requirement	Priority	Rel.	Ver.
FR-DLY-01	The system shall let staff mark each flower type as available today and enter the quantity in stems.	Must	R1	T
FR-DLY-02	The system shall show a "Today's flowers" section built from the flowers marked available today.	Must	R1	T
FR-DLY-03	The system shall show a badge such as "Available today only" on flowers flagged by staff as limited.	Should	R1	D
FR-DLY-04	The system shall hide a flower from ordering when its remaining quantity reaches zero.	Must	R1	T
FR-DLY-05	The system shall expire the "available today" status at the end of the business day (00:00 Damascus time) unless staff renew it.	Should	R1	T

3.2.3 Cart and checkout

ID	Requirement	Priority	Rel.	Ver.
FR-CRT-01	The system shall let a customer add, change the quantity of, and remove items in the cart.	Must	R1	T
FR-CRT-02	A designed bouquet shall be a single cart line that stores its bouquet spec, preview image and calculated price.	Must	R1	T
FR-CRT-03	The system shall keep the cart of a guest on the device and the cart of a signed-in customer across devices.	Should	R1	T
FR-CRT-04	The system shall let a customer order as a guest (name, phone, address) or with an account.	Must	R1	T
FR-CRT-05	The system shall let a customer add a gift card with a recipient name and a personal message of up to 200 characters.	Must	R1	T
FR-CRT-06	The system shall let a customer add order notes of up to 300 characters.	Must	R1	T
FR-CRT-07	The system shall let a customer apply a coupon code and shall show the discount before payment.	Should	R1	T
FR-CRT-08	The system shall show a price breakdown (items, extras, delivery fee, discount, total) in the selected currency before the order is placed.	Must	R1	T
FR-CRT-09	The system shall validate the phone number as an international number, accepting Syrian numbers in the	Must	R1	T

ID	Requirement	Priority	Rel.	Ver.
	forms 09XXXXXXXX and +9639XXXXXXXX.			
FR-CRT-10	The system shall prevent duplicate orders when the customer submits the checkout twice.	Must	R1	T
FR-CRT-11	The system shall offer the checkout as a single page on the website and as a short stepped flow in the app.	Should	R2	D
FR-CRT-12	The system shall let a customer ask the shop to confirm the order on WhatsApp instead of paying online, using a pre-filled message that contains the order number and summary.	Must	R1	T

3.2.4 Delivery

ID	Requirement	Priority	Rel.	Ver.
FR-DLV-01	The system shall accept delivery addresses inside Damascus only, and shall show the WhatsApp contact for addresses outside the supported zones.	Must	R1	T
FR-DLV-02	The admin shall define delivery zones, each with a fee and an estimated delivery time.	Must	R1	T
FR-DLV-03	The system shall let the customer choose Today, Tomorrow or any date up to 60 days ahead for occasions.	Must	R1	T
FR-DLV-04	The admin shall define delivery time slots with a capacity per slot, and the system shall stop offering a full slot.	Should	R1	T
FR-DLV-05	The admin shall set a minimum lead time per product type (for example 24 hours for custom cakes), and the system shall not offer earlier delivery times.	Must	R1	T
FR-DLV-06	The customer shall give the delivery address as a map pin plus free-text landmark, and may tick "Call before arrival".	Must	R1	T
FR-DLV-07	The customer shall give a recipient name and phone that differ from the buyer's, and may choose "Do not reveal the sender".	Should	R1	T
FR-DLV-08	The system shall offer pickup from the branch as an alternative to delivery.	Should	R1	T
FR-DLV-09	The system shall accept orders outside working hours	Must	R1	T

ID	Requirement	Priority	Rel.	Ver.
	(10:00 to 24:00) only for a delivery time inside working hours.			
FR-DLV-10	The system shall link to the Movo and BeeOrder pages of the shop as alternative ordering channels.	Could	R1	I

3.2.5 Payments

ID	Requirement	Priority	Rel.	Ver.
FR-PAY-01	The system shall support the payment methods cash on delivery, Sham Cash and Stripe, and the admin shall enable or disable each one.	Must	R1	T
FR-PAY-02	For cash on delivery, the system shall create the order as "Pending confirmation" until staff confirm it by phone or WhatsApp.	Must	R1	T
FR-PAY-03	For Sham Cash, the system shall mark the order paid automatically when the provider confirms the payment; where no automatic confirmation exists, the customer shall enter the transaction reference or upload a receipt image and staff shall verify it in the dashboard.	Must	R1	T
FR-PAY-04	For Stripe, the system shall redirect to Stripe Checkout, charge in USD, and mark the order paid only after a verified webhook.	Should	R3	T
FR-PAY-05	The system shall show prices in SYP or USD as chosen by the customer, using an exchange rate set by the admin.	Must	R1	T
FR-PAY-06	The system shall store on each order the currency, the exchange rate and the prices at the time of placement.	Must	R1	T
FR-PAY-07	The system shall access each payment method through a common payment-provider interface, so a method can be added or replaced without changing the checkout.	Must	R1	I
FR-PAY-08	The system shall never receive or store card numbers.	Must	R3	I
FR-PAY-09	The system shall keep a payment history per order (status, method, reference, amount, currency, time).	Must	R1	T
FR-PAY-10	The system shall cancel an unpaid online-payment order after a configurable time (default 2 hours) and	Should	R1	T

ID	Requirement	Priority	Rel.	Ver.
	release its reserved stock.			
FR-PAY-11	An authorized admin shall record a refund or a cancellation with a reason, and the system shall keep it in the order history.	Should	R1	T

3.2.6 Orders

ID	Requirement	Priority	Rel.	Ver.
FR-ORD-01	The system shall give each order a unique, short, human-readable number.	Must	R1	T
FR-ORD-02	An order shall move through the statuses Placed, Confirmed, In preparation, Ready, Out for delivery and Delivered, and may end as Cancelled.	Must	R1	T
FR-ORD-03	The system shall record each status change with the time and the user who made it.	Must	R1	T
FR-ORD-04	A customer shall see the order status in the account or, for guests, through the order number and phone number.	Must	R1	T
FR-ORD-05	A customer shall be able to cancel an order until it is Confirmed; after that the system shall direct the customer to WhatsApp.	Should	R1	T
FR-ORD-06	The system shall keep, on each order, a copy of the product names, prices, options and bouquet spec as they were at placement.	Must	R1	T
FR-ORD-07	The system shall let a signed-in customer reorder a previous order in one action.	Could	R2	D
FR-ORD-08	The dashboard shall alert staff in real time when a new order arrives.	Should	R1	D

3.2.7 Bouquet designer: common rules

مصمم الباقة هو أهم ميزة في المشروع. كل مراحله الثلاث (2D و3D وAI) تعمل من نفس الـ bouquet spec (الملحق B)، لذلك يمكن للزبون التنقل بينها دون إعادة التصميم.

ID	Requirement	Priority	Rel.	Ver.
FR-DSG-01	The designer shall guide the customer through the choices Occasion, Main flowers, Secondary flowers and greens, Color palette, Size, Wrap, Ribbon, Extras, and	Must	R1	D

ID	Requirement	Priority	Rel.	Ver.
	Card message.			
FR-DSG-02	The customer shall add and remove flowers with a quantity control, and the designer shall show each flower's unit price and availability.	Must	R1	T
FR-DSG-03	The designer shall offer only flowers whose stock allows the chosen quantity, and shall warn the customer and suggest a substitute if a chosen flower becomes unavailable before checkout.	Must	R1	T
FR-DSG-04	The designer shall offer the sizes Small, Medium, Large and Luxury, each with a minimum and maximum number of stems set by the admin.	Must	R1	T
FR-DSG-05	The designer shall offer color palettes (Pastel, Romantic red, Luxury white, Mixed) and shall highlight the flowers that match the chosen palette.	Should	R1	D
FR-DSG-06	The designer shall offer wraps (kraft, satin, transparent, box) and ribbon colors, each with its own price.	Must	R1	T
FR-DSG-07	The designer shall offer extras (chocolate, teddy bear, balloon, card, perfume, candle, mini cake) with their prices and stock.	Must	R1	T
FR-DSG-08	The designer shall recalculate and show the total price after every change.	Must	R1	T
FR-DSG-09	The designer shall produce a bouquet spec that validates against the schema in Appendix B.	Must	R1	T
FR-DSG-10	The customer shall be able to undo, redo and reset the design.	Should	R1	T
FR-DSG-11	The designer shall keep an unfinished design on the device and restore it when the customer returns.	Should	R1	T
FR-DSG-12	A signed-in customer shall be able to save designs under "My designs" and reopen them.	Should	R2	T
FR-DSG-13	The customer shall be able to add the design to the cart, or send the design summary and preview image to the shop on WhatsApp for confirmation.	Must	R1	T
FR-DSG-14	The admin shall define design rules (minimum stems, maximum stems, flowers that cannot be combined), and the designer shall enforce them with a clear	Should	R1	T

ID	Requirement	Priority	Rel.	Ver.
	message.			
FR-DSG-15	Every preview shall carry the notice "The final bouquet may differ slightly depending on the fresh flowers available".	Must	R1	I
FR-DSG-16	The system shall generate a bouquet recipe for each ordered design, listing stems per flower type, greens, wrap, ribbon, extras, card message, delivery time and the preview image, printable from the dashboard.	Must	R1	T
FR-DSG-17	The designer may show the estimated weight of the bouquet for handling and delivery.	Could	R3	A
FR-DSG-18	A customer may share a link to a saved design.	Could	R3	D

3.2.8 Designer stage 1: 2D layered preview (Smart Slots)

ID	Requirement	Priority	Rel.	Ver.
FR-D2D-01	The system shall render the bouquet preview by layering cut-out images in this order: wrap back, greens, flowers, wrap front, ribbon, extras.	Must	R1	D
FR-D2D-02	Each size shall have a layout of Smart Slots, each with position, scale, rotation and layer, and selected flowers shall be placed into the slots in a defined order.	Must	R1	T
FR-D2D-03	For the same bouquet spec, the 2D preview shall always look the same on every device and on every visit.	Must	R1	T
FR-D2D-04	The website and the app shall render the 2D preview from the same bouquet spec and the same slot layout data.	Must	R2	I
FR-D2D-05	The system shall export the 2D preview as an image for the cart, the order, the recipe and the WhatsApp message.	Must	R1	T
FR-D2D-06	The system shall show a neutral placeholder when a cut-out image fails to load, without blocking the designer.	Should	R1	T
FR-D2D-07	The admin shall upload and replace cut-out images for flowers, greens, wraps, ribbons and extras, and the system shall reject files that are not PNG or WebP with transparency.	Must	R1	T

ID	Requirement	Priority	Rel.	Ver.
FR-D2D-08	The admin may edit slot layouts in the dashboard.	Could	R3	D

3.2.9 Designer stage 2: interactive 3D preview

ID	Requirement	Priority	Rel.	Ver.
FR-D3D-01	The system shall show the bouquet as a 3D scene built from GLB models for each flower, green, wrap, ribbon and extra.	Should	R3	D
FR-D3D-02	The customer shall rotate the bouquet 360 degrees and zoom within limits, and the scene shall rotate slowly until the customer touches it.	Should	R3	D
FR-D3D-03	The customer shall add flowers, change quantity, colors and wrap while in 3D, and the same bouquet spec shall be updated.	Should	R3	T
FR-D3D-04	The customer shall switch between the 2D and 3D previews without losing the design.	Should	R3	T
FR-D3D-05	The 3D scene shall use soft lighting and shadows, and on phones it may respond lightly to device tilt.	Could	R3	D
FR-D3D-06	The system shall lower the 3D quality automatically on weak devices, and shall fall back to the 2D preview when 3D is not supported.	Must	R3	T
FR-D3D-07	The mobile app shall provide the same 3D preview as the website.	Should	R3	D
FR-D3D-08	The admin shall upload GLB models and the system shall validate format and file size before accepting them.	Should	R3	T

3.2.10 Designer stage 3: AI realistic preview

ID	Requirement	Priority	Rel.	Ver.
FR-DAI-01	The customer shall be able to press "Show the bouquet realistically" to generate an AI image of the designed bouquet.	Should	R3	D
FR-DAI-02	The server shall build the image prompt only from the structured bouquet spec (flower names and counts, greens, wrap, ribbon, lighting and background style), and shall never insert the card message or free-text notes into it.	Must	R3	I

ID	Requirement	Priority	Rel.	Ver.
FR-DAI-03	The generated image shall be labelled "AI-generated preview" with the notice in FR-DSG-15.	Must	R3	I
FR-DAI-04	The generation shall run as a background job with a progress indicator, a timeout of 60 seconds and one retry, and on failure the designer shall continue with the 2D or 3D preview.	Must	R3	T
FR-DAI-05	The system shall limit AI generations per customer or device per day (default 5) and shall stop all generations when a daily budget set by the admin is reached.	Must	R3	T
FR-DAI-06	The system shall cache the generated image by the hash of the bouquet spec and reuse it for identical designs.	Should	R3	T
FR-DAI-07	The customer shall be able to attach the generated image to the order, and the system shall store it with the order.	Should	R3	T
FR-DAI-08	Ordering shall never depend on the AI preview.	Must	R3	T
FR-DAI-09	The system shall log each generation request with the spec hash, the provider and the outcome, without personal data.	Should	R3	I

3.2.11 Cakes and sweets customization

ID	Requirement	Priority	Rel.	Ver.
FR-CKE-01	The customer shall choose, for a customizable cake, the size or number of servings, the flavor, the color or theme, and the design option defined by the admin.	Must	R1	T
FR-CKE-02	The customer shall be able to write a message to appear on the cake, of up to 40 characters, in Arabic or Latin letters.	Must	R1	T
FR-CKE-03	The system shall calculate the cake price from size, flavor and options, and shall apply the product's lead time.	Must	R1	T
FR-CKE-04	The customer may attach a reference photo of the desired design.	Could	R2	D
FR-CKE-05	The product page shall list the main allergens (such as nuts, gluten, milk, eggs) set by the admin.	Should	R1	I
FR-CKE-06	The system shall let a customer build a sweets or	Should	R1	T

ID	Requirement	Priority	Rel.	Ver.
	pastries box by choosing items and quantities up to the box capacity.			

3.2.12 Split box builder (flowers and sweets)

ID	Requirement	Priority	Rel.	Ver.
FR-BOX-01	The admin shall define box types, each with a number of compartments and the kind of item each compartment accepts (flowers or sweets).	Should	R1	T
FR-BOX-02	The customer shall choose a box type and fill each compartment with allowed items, and shall see a live preview.	Should	R1	D
FR-BOX-03	The system shall store the box as a box spec that follows the same rules as the bouquet spec and shall include it in the recipe.	Should	R1	T
FR-BOX-04	The system shall calculate the box price from the box, the items and the wrap, and shall apply the longest lead time among the items.	Should	R1	T

3.2.13 Customer accounts

ID	Requirement	Priority	Rel.	Ver.
FR-ACC-01	The system shall let a customer register and sign in with email and password, with Google, or with a phone number and OTP.	Must	R1	T
FR-ACC-02	The system shall let a customer reset a forgotten password by email or OTP.	Must	R1	T
FR-ACC-03	The customer shall edit name, phone, email and preferred language in the profile.	Must	R1	T
FR-ACC-04	The customer shall save several delivery addresses and mark one as default.	Should	R1	T
FR-ACC-05	The customer shall see past and current orders with status and details.	Must	R1	T
FR-ACC-06	The customer shall keep a favorites list of products and designs.	Could	R2	D
FR-ACC-07	The customer shall save occasions (person, type, date) and receive a reminder a configurable number of days before (default 7), with a shortcut to design or reorder	Should	R2	T

ID	Requirement	Priority	Rel.	Ver.
	a bouquet.			
FR-ACC-08	The customer shall be able to request deletion of the account and export of personal data.	Should	R2	D
FR-ACC-09	After a guest order, the system may offer to create an account linked to the same phone number.	Could	R2	D

3.2.14 Notifications and WhatsApp

ID	Requirement	Priority	Rel.	Ver.
FR-NTF-01	The system shall send the customer an order confirmation after placement.	Must	R1	T
FR-NTF-02	The system shall notify the customer when the order is Confirmed, Out for delivery and Delivered.	Should	R1	T
FR-NTF-03	The system shall notify the shop of each new order in the dashboard and by a WhatsApp message to the shop number.	Should	R1	T
FR-NTF-04	The system shall send push notifications to app users for order status changes and occasion reminders.	Must	R2	T
FR-NTF-05	The customer shall be able to opt out of marketing and reminder messages, while order messages remain active.	Must	R1	T
FR-NTF-06	The admin shall edit message templates in Arabic and English.	Could	R3	D
FR-NTF-07	The system shall retry a failed notification and log the outcome.	Should	R1	T

3.2.15 Inventory

ID	Requirement	Priority	Rel.	Ver.
FR-INV-01	The system shall track stock for each flower (in stems), green, wrap, ribbon, extra and ready item, and shall allow made-to-order items without stock.	Must	R1	T
FR-INV-02	The system shall reserve stock when an order is placed and release it when the order is cancelled or stays unconfirmed beyond a configurable time.	Must	R1	T
FR-INV-03	The system shall prevent selling more than the available stock when several customers order at the same time.	Must	R1	T

ID	Requirement	Priority	Rel.	Ver.
FR-INV-04	The admin shall set a low-stock threshold per item, and the dashboard shall alert staff when stock falls below it.	Should	R1	T
FR-INV-05	Staff shall adjust stock manually with a reason, and the system shall log the change.	Must	R1	T
FR-INV-06	The admin shall define seasonal availability windows for items, and the system shall hide an item outside its window.	Should	R1	T

3.2.16 Admin dashboard

ID	Requirement	Priority	Rel.	Ver.
FR-ADM-01	The dashboard shall support the roles Owner, Manager, Florist and Delivery coordinator, each with its own permissions.	Must	R1	T
FR-ADM-02	Authorized staff shall create, edit, hide and delete products with images, categories, options, prices and lead times, in Arabic and English.	Must	R1	T
FR-ADM-03	Authorized staff shall manage the designer items (flowers, greens, wraps, ribbons, extras): names, prices, stock, color and occasion tags, cut-out images and 3D models.	Must	R1	T
FR-ADM-04	Authorized staff shall list, filter and search orders, change their status, assign a delivery person, add internal notes, and print the order slip and the bouquet recipe.	Must	R1	T
FR-ADM-05	The Owner or Manager shall set prices and the exchange rate between SYP and USD, and may update several prices at once.	Must	R1	T
FR-ADM-06	The Owner or Manager shall create coupons (percentage or fixed amount) with validity dates, minimum order, usage limit and eligible categories.	Should	R1	T
FR-ADM-07	The Owner or Manager shall set delivery zones, time slots, lead times, working hours and closed days.	Must	R1	T
FR-ADM-08	The Owner or Manager shall edit home page banners, hero media, occasion pages and policy pages.	Should	R1	D
FR-ADM-09	The system shall keep an audit log of changes to prices, stock, order status, coupons and user roles, showing	Must	R1	T

ID	Requirement	Priority	Rel.	Ver.
	who changed what and when.			
FR-ADM-10	The dashboard home shall show today's orders, pending confirmations, today's revenue and low-stock items.	Should	R1	D
FR-ADM-11	The Owner and Manager accounts shall use a second sign-in factor.	Should	R1	T
FR-ADM-12	The Owner or Manager shall switch the shop to "busy" or "closed" mode, which pauses new orders and shows a message to customers.	Should	R1	T
FR-ADM-13	Authorized staff may import and export products as CSV files.	Could	R3	D

3.2.17 Reports

ID	Requirement	Priority	Rel.	Ver.
FR-RPT-01	The system shall report sales by day, week and month, by category, by product, by payment method and by delivery zone.	Should	R1	T
FR-RPT-02	The system shall report designer usage: designs started and completed, drop-off by step, most used flowers, average stems, and use of 2D, 3D and AI preview.	Should	R3	T
FR-RPT-03	The system shall show money amounts in SYP and USD.	Should	R1	T
FR-RPT-04	The system shall report coupon usage.	Could	R3	T

3.2.18 Content, reviews and branch information

ID	Requirement	Priority	Rel.	Ver.
FR-CON-01	The system shall let customers rate a delivered order from 1 to 5 stars with a comment, and shall publish it only after admin approval.	Should	R3	T
FR-CON-02	The home page shall show a gallery of Instagram posts and Reels from @bellebiens.	Could	R3	D
FR-CON-03	The branch page shall show the map, phone numbers, WhatsApp, Instagram link and working hours (10:00 to 24:00, editable by the admin).	Must	R1	I
FR-CON-04	The system shall provide pages for delivery policy, cancellation policy, privacy policy and terms of use, in	Must	R1	I

ID	Requirement	Priority	Rel.	Ver.
	Arabic and English.			
FR-CON-05	The hero video shall start muted, load after the first screen is shown, and be replaced by an image on slow connections.	Should	R1	T

3.2.19 Local SEO (website)

ID	Requirement	Priority	Rel.	Ver.
FR-SEO-01	Website pages shall be rendered on the server so search engines receive full content, with a unique title and description per page in Arabic and English.	Must	R1	I
FR-SEO-02	The website shall provide sitemap.xml, robots.txt, canonical links and hreflang links for Arabic and English.	Must	R1	I
FR-SEO-03	The website shall provide structured data for the local business, products and breadcrumbs.	Should	R1	I
FR-SEO-04	Page titles, headings and content shall target local searches such as "كيك", "توصيل ورد دمشق", "ورد دمشق" and "حلويات أبو رمانة دمشق".	Should	R1	I
FR-SEO-05	Shared links (Instagram, WhatsApp) shall show a preview image, title and description through Open Graph tags.	Must	R1	T
FR-SEO-06	Staff may edit the title and description of a page from the dashboard.	Could	R3	D

3.2.20 Localization and analytics

ID	Requirement	Priority	Rel.	Ver.
FR-LOC-01	The system shall provide full RTL layout for Arabic, including mirrored navigation and icons.	Must	R1	I
FR-LOC-02	The system shall show Western digits by default and Arabic-Indic digits as a user setting.	Should	R1	T
FR-LOC-03	The system shall store and show times in the Asia/Damascus time zone and dates in the Gregorian calendar.	Must	R1	T
FR-LOC-04	Every customer-facing text field managed in the dashboard shall have an Arabic and an English value, and the system shall show the other language when	Must	R1	T

ID	Requirement	Priority	Rel.	Ver.
	one is missing.			
FR-ANL-01	The system shall ask for consent before using analytics or marketing trackers on the website and in the app.	Should	R1	T
FR-ANL-02	The system shall record the events product view, designer started, designer completed, add to cart, checkout started and purchase.	Should	R1	T

3.3 Use cases

These five flows cover the main value of the platform. Each lists the requirements it exercises, for tracing in Appendix A.

UC-01 Design and order a custom bouquet

Field	Content
Actor	Customer in Damascus
Related requirements	FR-DSG-01 to 16, FR-D2D-01 to 05, FR-CRT-01 to 08, FR-DLV-03, FR-PAY-02
Preconditions	Designer items, cut-out images and stock are loaded; the shop is open for orders
Main flow	1. The customer taps "Design your bouquet". 2. The customer chooses the occasion, main flowers, greens, palette, size, wrap, ribbon and extras. 3. The system updates the preview and the price after each choice. 4. The customer writes a card message. 5. The customer taps "Add to cart". 6. The customer chooses delivery date, time slot and address, and pays on delivery. 7. The system creates the order, reserves the stock and notifies the shop.
Alternative flows	3a. A chosen flower runs out: the system warns and suggests a substitute (FR-DSG-03). 5a. The customer taps "Send on WhatsApp" instead: the system opens WhatsApp with the preview and the summary (FR-DSG-13). 6a. The address is outside Damascus: the system refuses and shows the WhatsApp contact (FR-DLV-01).
Postconditions	An order in status Placed holds the bouquet spec and the preview image; stock is reserved

UC-02 Order a customized cake and pay with Sham Cash

Field	Content

Field	Content
Actor	Customer in Damascus
Related requirements	FR-CKE-01 to 03, FR-DLV-05, FR-PAY-03, FR-PAY-05, FR-PAY-06, FR-ORD-02
Preconditions	The cake product has sizes, flavors and a lead time
Main flow	1. The customer opens a cake and picks size, flavor, theme and message. 2. The system shows the earliest delivery time based on the lead time. 3. The customer chooses a date and pays with Sham Cash. 4. The system shows the payment instructions and the amount in the chosen currency. 5. The customer pays and enters the reference or uploads the receipt. 6. Staff verify the payment in the dashboard and confirm the order. 7. The customer receives a confirmation.
Alternative flows	5a. The provider confirms automatically: the order is marked paid without staff action. 6a. Staff reject the receipt: the customer is notified and may retry or switch to cash on delivery. 6b. The customer does not pay within the time limit: the order is cancelled and stock released (FR-PAY-10).
Postconditions	A paid and confirmed order with the exchange rate stored

UC-03 Update today's flowers

Field	Content
Actor	Florist
Related requirements	FR-DLY-01 to 05, FR-INV-01, FR-INV-05, FR-ADM-03
Preconditions	The florist is signed in to the dashboard
Main flow	1. The florist opens "Today's flowers". 2. The florist marks each flower available and enters the stem count. 3. The florist flags limited flowers. 4. The system publishes the list on the website and in the app and enables those flowers in the designer.
Alternative flows	4a. A flower is left at zero: it stays hidden from ordering.
Postconditions	The customer-facing availability matches the shop stock

UC-04 Prepare and deliver an order from the recipe

Field	Content
Actor	Florist, delivery coordinator

Field	Content
Related requirements	FR-ORD-02, FR-ORD-03, FR-ORD-08, FR-DSG-16, FR-NTF-02, EI-HW-01
Preconditions	An order exists in status Placed
Main flow	1. The dashboard alerts staff of the new order. 2. Staff confirm the order by call or WhatsApp. 3. The florist prints the recipe and the order slip. 4. The florist sets In preparation, then Ready. 5. The coordinator assigns a delivery person and sets Out for delivery. 6. After delivery the coordinator sets Delivered. 7. The customer is notified at each step.
Alternative flows	2a. The customer does not answer: staff cancel with a reason and stock is released. 6a. Delivery fails: staff add a note and contact the customer.
Postconditions	The order is Delivered and every change is in the audit log

UC-05 Send a gift from abroad and pay by card

Field	Content
Actor	Customer abroad
Related requirements	FR-PAY-04, FR-PAY-05, FR-PAY-08, FR-DLV-07, FR-DAI-01 to 03
Preconditions	Stripe is enabled and legally available (risk R-01)
Main flow	1. The customer designs a bouquet and previews it in AI. 2. The customer enters the recipient's name, phone and Damascus address and ticks "Do not reveal the sender". 3. The customer pays by card in USD on Stripe Checkout. 4. Stripe confirms through a webhook and the order becomes paid. 5. The recipient receives the bouquet and the sender is notified of the delivery.
Alternative flows	3a. The payment fails: the order stays unpaid and the customer can retry.
Postconditions	A paid order in USD; no card data stored

4. Non-functional (quality) requirements

Quality requirements follow the ISO/IEC 25010 characteristics, each with a number a test can check. الأرقام هي أهداف أولية اقترحتها ويجب تأكيدها مع العميل (OI-03). Priorities apply to all releases unless stated.

ID	Characteristic	Requirement	Priority	Ver.
NFR-PERF-01	Performance	The website home page and product	Must	T

ID	Characteristic	Requirement	Priority	Ver.
	efficiency	pages shall reach Largest Contentful Paint of 2.5 s or less at the 75th percentile on the Lighthouse mobile profile (slow 4G).		
NFR-PERF-02	Performance efficiency	The JavaScript loaded for the home page shall not exceed 200 KB compressed, and the designer code shall load only when the designer opens.	Should	I
NFR-PERF-03	Performance efficiency	The 2D preview and the price shall update within 200 ms of a customer change on a mid-range Android phone from 2020 or later.	Must	T
NFR-PERF-04	Performance efficiency	The 3D preview shall run at 30 frames per second or more on a mid-range phone, or lower its quality automatically until it does.	Should	T
NFR-PERF-05	Performance efficiency	Each flower GLB model shall not exceed 500 KB after compression, and the first 3D scene shall not download more than 3 MB.	Should	I
NFR-PERF-06	Performance efficiency	The API shall answer 95% of requests within 500 ms with 100 concurrent users, excluding AI generation.	Must	T
NFR-PERF-07	Performance efficiency	Product images shall be served in WebP or AVIF through the CDN in sizes matched to the screen, with a product card image of 150 KB or less on mobile.	Must	I
NFR-PERF-08	Performance efficiency	The system shall support 500 concurrent visitors and 50 orders per hour during peak days without breaching NFR-PERF-06.	Should	T
NFR-REL-01	Reliability	The system shall be available 99.5% of the time per calendar month, excluding planned maintenance announced 24 hours ahead.	Should	A
NFR-REL-02	Reliability	The database shall be backed up daily and kept 30 days, with a recovery point of 24 hours and a recovery time of 4 hours, and a restore shall be tested each quarter.	Must	D

ID	Characteristic	Requirement	Priority	Ver.
NFR-REL-03	Reliability	The system shall store an order before telling the customer it succeeded, and an order shall stay visible in the dashboard even if every notification fails.	Must	T
NFR-REL-04	Reliability	If the AI, 3D, payment or messaging service is down, the rest of the platform shall keep working and show a clear message for the affected feature.	Must	T
NFR-REL-05	Reliability	The mobile app shall retry failed requests and show the last loaded catalog when the connection drops.	Should	T
NFR-SEC-01	Security	The server shall check the user's role on every API call, and the clients shall not be trusted to enforce permissions.	Must	T
NFR-SEC-02	Security	Passwords shall be stored only as salted hashes (Argon2id or bcrypt) and shall have at least 8 characters.	Must	I
NFR-SEC-03	Security	The system shall rate-limit sign-in, OTP requests, coupon attempts and AI generations.	Must	T
NFR-SEC-04	Security	The system shall protect against the OWASP Top 10 risks, run automatic dependency scanning, and pass a security review before each release.	Must	I
NFR-SEC-05	Security	API keys for payments, messaging, AI and media shall exist only on the server or in a secret store, never in website or app code.	Must	I
NFR-SEC-06	Security	The server shall calculate every price, discount and total itself and shall ignore prices sent by a client.	Must	T
NFR-SEC-07	Security	The server shall validate each bouquet spec and box spec against its schema and its design rules before accepting an order.	Must	T
NFR-SEC-08	Security	Uploaded files (receipts, reference photos, cut-outs, 3D models) shall be checked for type, size and content, and shall be re-	Must	T

ID	Characteristic	Requirement	Priority	Ver.
		encoded or scanned before use.		
NFR-SEC-09	Security	Personal data (name, phone, address) shall be encrypted in transit and at rest, including backups.	Must	I
NFR-SEC-10	Security	OTP codes shall expire after 5 minutes and allow at most 5 attempts.	Must	T
NFR-SEC-11	Security	Dashboard sessions shall end after 30 minutes of inactivity.	Should	T
NFR-SEC-12	Security	The audit log shall be append-only for all users, including administrators.	Should	I
NFR-INT-01	Interaction capability	At least 80% of first-time test users shall complete a designed-bouquet order in 5 minutes or less without help, in a test with at least 5 users.	Must	T
NFR-INT-02	Interaction capability	The website and the dashboard shall meet WCAG 2.1 level AA, and the app shall provide screen-reader labels and support larger text sizes.	Should	I
NFR-INT-03	Interaction capability	Touch targets shall be at least 44 by 44 px (48 dp in the app).	Must	I
NFR-INT-04	Interaction capability	The designer and checkout shall work on a 360 px wide screen without horizontal scrolling.	Must	T
NFR-INT-05	Interaction capability	Error messages shall be in the customer's language and say what to do next.	Must	I
NFR-INT-06	Interaction capability	The designer may show a short first-use guide that the customer can dismiss.	Could	D
NFR-MNT-01	Maintainability	Automated tests shall cover at least 70% of backend business logic, and at least 90% of pricing, stock and delivery rules.	Must	A
NFR-MNT-02	Maintainability	A pipeline shall run lint, tests and build on every change and shall deploy releases to a staging environment before production.	Must	I
NFR-MNT-03	Maintainability	The bouquet spec schema and the pricing rules shall be defined once and verified	Must	T

ID	Characteristic	Requirement	Priority	Ver.
		with shared test cases on the server, the website and the app.		
NFR-MNT-04	Maintainability	The OpenAPI document shall always match the deployed API.	Should	I
NFR-MNT-05	Maintainability	The system shall write structured logs and send errors to a monitoring service with alerts to the technical lead.	Should	D
NFR-MNT-06	Maintainability	The project shall deliver a README, an architecture note, an operations runbook and an Arabic user guide for the dashboard.	Must	I
NFR-FLX-01	Flexibility	The website shall run on the browsers and phones listed in 2.4, and the app on Android 9+ and iOS 15+ from one Flutter codebase.	Must	T
NFR-FLX-02	Flexibility	Payment, messaging, AI and media services shall each sit behind an interface so a provider can be replaced without changing business logic.	Must	I
NFR-FLX-03	Flexibility	The data model shall keep a branch identifier on orders and stock, so more branches or cities can be added later.	Could	I
NFR-CMP-01	Compatibility	The backend shall keep working for app versions up to two releases old, and the app shall show an update screen when its version is no longer supported.	Should	T
NFR-CMP-02	Compatibility	The dashboard shall export and import data in CSV (UTF-8 with Arabic text) that opens correctly in Excel.	Should	T
NFR-FUN-01	Functional suitability	The price of a product, bouquet, cake or box shall be the same on the server, the website and the app for the same inputs.	Must	T
NFR-FUN-02	Functional suitability	Stock shall never become negative, and delivery fees, lead times and slots shall always follow the admin settings.	Must	T
NFR-SAF-01	Safety	Not applicable: the system cannot cause	n/a	I

ID	Characteristic	Requirement	Priority	Ver.
		physical harm. Allergen information is covered by FR-CKE-05.		

5. Data, constraints, legal and content requirements

This section covers what the system stores, what it is forced to use, which outside rules apply, and which real-world assets the shop must supply.

5.1 Data requirements

ID	Entity	Key attributes	Rules
DR-01	User	id, name, phone, email, language, role, status	Phone unique per account; role from FR-ADM-01
DR-02	Address	user, label, zone, map point, landmark, call-before-arrival	A user may have several; one default
DR-03	Product	id, category, names and descriptions (ar, en), images, base price, lead time, allergens, status	Names and descriptions required in both languages
DR-04	Product option	product, type (size, flavor, theme), name, price change	Used by cakes and sweets
DR-05	Designer item	id, type (flower, green, wrap, ribbon, extra), names (ar, en), unit price, color tags, occasion tags, cut-out image, GLB model, season window	Type decides which designer step shows it
DR-06	Slot layout	size, list of slots (x, y, scale, rotation, layer)	One layout per size, versioned
DR-07	Box type	name, compartments, accepted kinds per compartment	Used by FR-BOX-01
DR-08	Bouquet design	owner or device, bouquet spec (JSON), preview image, schema version, spec hash	Immutable once ordered
DR-09	Order	number, customer or guest, recipient, delivery date and slot, zone, status, currency, exchange rate, totals, notes, gift card	Totals computed by the server
DR-10	Order line	order, product or design, quantity, price snapshot, option snapshot	Copy of data at placement (FR-ORD-06)

ID	Entity	Key attributes	Rules
DR-11	Payment	order, method, status, reference, amount, currency, receipt file, time	History kept (FR-PAY-09)
DR-12	Stock level and movement	item, quantity, reserved, movement type, reason, user, time	Quantity never below zero
DR-13	Daily availability	flower, date, quantity, limited flag	Expires at end of day
DR-14	Delivery zone, time slot	zone, fee, estimated time; slot, capacity, days	Edited by Owner or Manager
DR-15	Coupon	code, type, value, validity, minimum order, limits, categories	Code unique
DR-16	AI preview	spec hash, image, provider, created time, linked order	Cached by hash (FR-DAI-06)
DR-17	Review	order, rating, comment, status	Published only when approved
DR-18	Occasion reminder	user, person, type, date, days before	Used by FR-ACC-07
DR-19	Notification log	recipient, channel, template, status, time	Kept for retries and audit
DR-20	Audit log entry	user, action, object, old and new value, time	Append-only
DR-21	Setting	exchange rate, working hours, closed days, shop mode, limits	Changes are audited

5.2 Data retention (proposal)

- Orders and payments: kept for the period required by the shop's accounting rules; default 5 years (OI-06).
- Unused AI previews: deleted after 90 days; previews attached to an order stay with the order.
- Payment receipt images and reference photos: deleted 12 months after delivery.
- Notification and error logs: kept 90 days.
- OTP codes: deleted after use or expiry.

5.3 Design and implementation constraints

These choices were made in the analysis conversation or by me to fit it. The technical lead may change one with the client's approval. المبررات في الملحق D.

ID	Constraint	Reason
DC-01	All three clients shall use one shared backend REST API.	One set of rules for price, stock and orders
DC-02	The website and the dashboard shall be built with Next.js, React and TypeScript.	Server rendering for SEO, fast mobile pages
DC-03	The mobile app shall be built with Flutter.	One codebase for Android and iOS
DC-04	The backend shall be a NestJS (TypeScript) service on PostgreSQL, with Prisma as the data layer.	Shared language with the website, strong data integrity for stock and orders
DC-05	The 3D preview shall use Three.js with React Three Fiber and GLB models compressed with Draco.	Chosen in the analysis as the best fit
DC-06	Images, video and 3D models shall be stored in Cloudinary or an S3-compatible store behind a CDN.	Fast delivery of heavy media
DC-07	Sign-in shall be issued by the backend (session for the website, token for the app) with email, Google and phone OTP.	One account across clients
DC-08	Source code shall be in a Git repository with an automated pipeline.	NFR-MNT-02

5.4 Legal and compliance

This SRS is not legal advice; the owner should confirm these points with a lawyer before launch.

- LEG-01: The platform shall publish a privacy policy and terms of use, and shall record consent for analytics and marketing.
- LEG-02: Before choosing providers for payments (Stripe, Sham Cash), hosting, messaging (WhatsApp Business, SMS) and AI image generation, the owner shall confirm that each can lawfully serve a business in Syria and customers abroad. See risk R-01.
- LEG-03: The shop shall accept the merchant terms of each payment provider.
- LEG-04: Fonts, photos, 3D models, icons and libraries shall have licenses that allow commercial use.
- LEG-05: Allergen and ingredient information shall follow the local labeling rules for food, to be confirmed (OI-07).
- LEG-06: If customers in the EU or UK are served, GDPR or UK GDPR may apply to their data; this shall be checked (OI-08).
- LEG-07: Instagram content embedded on the site shall follow Instagram's terms and the shop's rights to the media.

5.5 3D asset requirements

ID	Requirement
DA-01	Each 3D model shall be a GLB file with the origin at the base of the stem, real-world scale, and PBR materials.
DA-02	Each model shall be at most 500 KB after Draco compression, with textures of at most 1024 px.
DA-03	File names shall follow type_name_variant_version, for example flower_red_rose_v1.glb, and match the designer item id.
DA-04	Each flower model shall exist in at least two color variants only when the real flower comes in them; other colors shall be set through materials.

5.6 Photo-shoot content requirements

The designer cannot launch without real images. هذه القائمة مأخوذة من توصيات المحادثة وتصوّر في محل Belle Biens نفسه.

ID	Content	Quantity	Specification
CT-01	Main flower types (cut-outs)	10 to 15 types	Same camera angle and light for all; three sizes per type (small, medium, large); transparent background
CT-02	Greens (eucalyptus, gypsophila, ruscus and similar)	3 to 5 types	Same rules as CT-01
CT-03	Wraps	5 types	Front and back views, separately
CT-04	Ribbons	4 colors	Photographed separately
CT-05	Common extras (teddy bear, chocolate, balloon, bento cake, matcha)	5 items	Same rules as CT-01
CT-06	Ready-made bouquets	10 bouquets	Reference for the shop and for the catalog
CT-07	Short videos for the hero and Reels	As needed	Vertical and horizontal versions, short, with no required sound
CT-08	Technical format for all cut-outs	All	At least 2000 px on the longest side, PNG or WebP with transparency, sRGB, no baked shadows (shadows are added by the app)
CT-09	Consistency	All	Fixed camera and fixed lighting for the whole session; a gray card in the first shot of each

ID	Content	Quantity	Specification
			session

6. Verification and acceptance

Every requirement carries a verification method, and each release is accepted only when its Must requirements pass.

6.1 Verification methods and test levels

Code	Method	Used for
T	Test	Behavior that can be run: unit, integration, end-to-end, performance, security, usability tests
I	Inspection	Code, configuration, documents, screens checked by review
A	Analysis	Reports, coverage figures, monitoring data, calculations
D	Demonstration	A live walk-through with the client in staging

Test level	Scope
Unit	Pricing, stock, delivery rules, bouquet spec validation, slot layout
Integration	API with database, payment and messaging providers (sandbox), webhooks
End-to-end	The five use cases UC-01 to UC-05 on website and app
Performance	NFR-PERF targets with a load tool and Lighthouse; 3D frame rate on three real phones
Security	OWASP checks, role tests, upload tests, rate limits
Usability	At least 5 first-time users on phones (NFR-INT-01)
Device and browser	Matrix from 2.4, including one low-end Android phone
User acceptance	Shop owner and staff run real scenarios in staging

6.2 Acceptance criteria for key requirements

Req. ID	Acceptance criterion	Result
FR-DSG-03	Given 6 white gypsophila stems in stock, when a customer selects 8, then the system blocks the extra stems and shows the available quantity.	Not run

Req. ID	Acceptance criterion	Result
FR-DSG-08, NFR-FUN-01	Given the same bouquet spec, when priced on the server, the website and the app, then the three totals are identical to the last digit.	Not run
FR-D2D-03	Given one bouquet spec, when rendered twice on two different phones, then the slot positions of every flower are identical.	Not run
FR-DLV-05	Given a custom cake with a 24-hour lead time ordered at 15:00, when the customer picks a delivery time, then no time before 15:00 the next day is offered.	Not run
FR-DLV-01	Given an address outside the supported zones, when the customer continues, then the order is refused and the WhatsApp contact is shown.	Not run
FR-PAY-03	Given a Sham Cash order without automatic confirmation, when staff approve the receipt, then the order becomes paid and the customer is notified.	Not run
FR-PAY-06	Given an order placed at a certain exchange rate, when the admin changes the rate afterwards, then the order total does not change.	Not run
FR-INV-03	Given 1 remaining item and two customers ordering at the same moment, when both submit, then exactly one order succeeds.	Not run
FR-DAI-02	Given a card message with the text of an instruction, when the AI preview is generated, then the prompt sent to the provider contains no part of the message.	Not run
FR-DAI-05	Given the daily AI budget is reached, when a customer asks for a preview, then the request is declined with a clear message and the designer keeps working.	Not run
NFR-SEC-06	Given a modified request carrying a lower price, when the order is submitted, then the server uses its own price.	Not run
NFR-PERF-01	Given the Lighthouse mobile profile, when the home page is tested, then LCP is 2.5 s or less at the 75th percentile.	Not run
FR-LOC-01	Given Arabic is selected, when any customer page is opened, then layout, navigation and icons are mirrored and no text is cut.	Not run

6.3 Release acceptance

- A release is accepted when all Must requirements assigned to it pass, no critical or high defect is open, and the shop owner signs the acceptance in the approvals table.
- Should and Could requirements that did not pass move to the next release or to the open issues list.

6.4 Change control

A change to an approved requirement needs a written change request that states the reason, the affected IDs, the cost and the schedule effect. The Owner approves it, the version in Document control is raised, and the traceability matrix is updated.

Appendix A: Requirements traceability matrix

Each business need from the analysis conversation maps to its requirements, use cases and tests.

Need	Business need	Requirements	Use case	Test level
N-01	Turn Instagram visitors into orders, mostly on phones	FR-CAT-09, FR-CAT-10, FR-SEO-01 to 06, FR-CON-05, NFR-PERF-01 to 07, EI-UI-02, EI-UI-04	UC-01	Performance, E2E
N-02	Let customers design their own bouquet and see it before ordering	FR-DSG-01 to 18, FR-D2D-01 to 08, FR-D3D-01 to 08, FR-DAI-01 to 09, FR-CAT-07	UC-01, UC-05	Unit, E2E, usability
N-03	Sell cakes, sweets boxes, gifts and flower-and-sweets boxes	FR-CAT-01 to 03, FR-CKE-01 to 06, FR-BOX-01 to 04	UC-02	E2E
N-04	Take orders and deliver inside Damascus on a chosen date	FR-CRT-01 to 12, FR-DLV-01 to 10, FR-ORD-01 to 08	UC-01, UC-02, UC-04	Unit, E2E
N-05	Accept payment from customers in Syria and abroad	FR-PAY-01 to 11, EI-SW-03, EI-SW-04, NFR-SEC-06	UC-02, UC-05	Integration, security
N-06	Cut the back-and-forth on WhatsApp	FR-CRT-12, FR-DSG-13, FR-DSG-16, FR-NTF-01 to 07, EI-SW-01, EI-SW-02	UC-01, UC-04	E2E
N-07	Run the shop day to day	FR-DLY-01 to 05, FR-INV-01 to 06, FR-ADM-01 to 13, FR-RPT-01 to 04	UC-03, UC-04	E2E, acceptance

Need	Business need	Requirements	Use case	Test level
N-08	A warm Arabic and English brand experience	EI-UI-01, EI-UI-05, FR-LOC-01 to 04, NFR-INT-01 to 06	UC-01	Usability, inspection
N-09	Build trust with new customers	FR-CON-01 to 04, FR-CAT-05	UC-01	Demonstration
N-10	A secure, reliable platform	NFR-SEC-01 to 12, NFR-REL-01 to 05, NFR-MNT-01 to 06, FR-ADM-09	All	Security, analysis

Appendix B: Bouquet spec (JSON schema, version 1)

The bouquet spec is the contract between the designer, the server, the recipe and the AI prompt. It carries what was chosen, not how it is drawn, so the same spec works in 2D, 3D and AI. Delivery date and address belong to the order, not to the spec. تم تحديث مثال المحادثة بإضافة schemaVersion وsize وpalette وكمية الإضافات.

```
{
  "schemaVersion": 1,
  "occasion": "birthday",
  "size": "medium",
  "palette": "romantic_red",
  "flowers": [
    { "id": "red_rose", "qty": 12 },
    { "id": "white_gypsophila", "qty": 6 }
  ],
  "greens": [ { "id": "eucalyptus", "qty": 3 } ],
  "wrap": "kraft_beige",
  "ribbon": "satin_pink",
  "extras": [
    { "id": "teddy_bear_small", "qty": 1 },
    { "id": "chocolate_box", "qty": 1 }
  ],
  "cardMessage": "كل عام وأنتِ أجمل",
  "previewMode": "3d"
}
```

Field	Type	Rule
schemaVersion	integer	Required; the server rejects unknown versions
occasion	text	One of the occasions defined by the admin
size	text	small, medium, large or luxury; stem count must stay inside the size limits (FR-DSG-04)
palette	text	Optional; one of the palettes in FR-DSG-05
flowers	list	Each id must be a flower item in stock; qty is a whole number of 1 or more; at least one flower

Field	Type	Rule
greens	list	Optional; each id must be a green item
wrap, ribbon	text	Must be ids of wrap and ribbon items
extras	list	Optional; each id must be an extra item with stock
cardMessage	text	Up to 200 characters; never sent to the AI provider (FR-DAI-02)
previewMode	text	2d, 3d or ai; affects display only, never the price

A box spec follows the same rules, with boxType and a list of compartments, each holding items allowed by the box type.

Appendix C: Risk register

ID	Risk	Likelihood	Impact	Mitigation
R-01	Stripe, some cloud, messaging or AI providers may not lawfully serve a business in Syria, or may refuse the account	High	High	Check availability and legal advice before the build; keep every provider behind an interface (NFR-FLX-02); launch with cash on delivery and Sham Cash; treat Stripe as a later, optional release
R-02	The shop delivers the photo content late or at low quality	High	High	Agree the shoot plan in 5.6 early; start with the minimum set; use placeholder art in staging
R-03	Sham Cash offers no merchant API or callback	Medium	High	Manual verification path is already a requirement (FR-PAY-03)
R-04	Scope is large: two clients and three designer stages	High	High	Release plan in 2.7; shared backend and shared spec; Must requirements first
R-05	3D preview in the Flutter app is slow or hard to build	Medium	Medium	Prototype early (decision D-05); fall back to 2D on weak devices (FR-D3D-06)
R-06	AI cost grows or outputs look unlike the real bouquet	Medium	Medium	Daily limits and budget (FR-DAI-05), caching, clear disclaimer (FR-DSG-15, FR-DAI-03)
R-07	Exchange-rate swings make prices wrong	Medium	High	Admin-controlled rate, price locked per order (FR-PAY-05, FR-PAY-06)
R-08	Staff do not update daily	Medium	High	Simple one-screen update (UC-03),

ID	Risk	Likelihood	Impact	Mitigation
	stock			expiry of availability (FR-DLY-05), low-stock alerts
R-09	Poor networks and low-end phones slow the site	High	Medium	Performance budgets (NFR-PERF), lazy loading, 2D fallback, image optimization
R-10	WhatsApp Business access is limited or blocked	Medium	Medium	wa.me hand-off and SMS fallback (EI-SW-01, EI-SW-02)

Appendix D: Decision log

The first rows record the choices you made when I asked; the rest record choices I made, with the reason, so you can overrule any of them.

ID	Decision	Reason	Decided by
D-01	Build a website (PWA), a Flutter app and a web admin dashboard	Instagram traffic needs a fast website with SEO; the shop also wants an app	Client
D-02	Use one shared NestJS REST backend instead of Next.js API routes	With two customer clients, one backend keeps price, stock and order rules identical; the conversation itself says to move to NestJS when the system grows	Analyst
D-03	All three designer stages are in v1.0, delivered in order 2D, 3D, AI	Client request; 2D comes first because 3D and AI use the same spec and 2D is the fallback	Client and analyst
D-04	Payments: cash on delivery, Sham Cash, Stripe; Sham Cash has a manual verification fallback; Stripe comes last	Client request; Stripe and Sham Cash availability is uncertain (R-01, R-03)	Client and analyst
D-05	3D in the Flutter app will first be tried by reusing the web 3D viewer inside the app; a native viewer is the alternative	Reuses the same GLB models and code; to be confirmed by a prototype before R3	Analyst
D-06	Requirements in English, explanations in Arabic	Client request	Client
D-07	The bouquet spec stores choices, not	Prevents price tampering and keeps	Analyst

ID	Decision	Reason	Decided by
	prices; the server prices it	all clients consistent (NFR-SEC-06)	
D-08	Prices can be shown in SYP and USD; each order locks the rate	Exchange rates move quickly (R-07)	Analyst
D-09	The AI prompt is built only from the spec	Blocks prompt injection through free text (FR-DAI-02)	Analyst
D-10	A custom build, not a ready-made Shopify theme	The conversation concluded that the designer needs real customization	Analysis conversation
D-11	Instagram, Movo and BeeOrder stay as channels linked from the site, not integrations	Keeps v1.0 focused; the shop already uses them	Analyst

Requirements I added beyond the conversation: customer accounts and privacy (FR-ACC-08), delivery slots, lead times and recipient options (FR-DLV-04, 05, 07), unpaid-order expiry and refunds (FR-PAY-10, 11), duplicate-order protection (FR-CRT-10), stock reservation and concurrency (FR-INV-02, 03), audit log, 2FA and busy mode (FR-ADM-09, 11, 12), reports (FR-RPT), notification retries, consent for analytics, the security, reliability and maintainability sets in Section 4, the legal list in 5.4, data retention in 5.2, and change control in 6.4.

Appendix E: Open issues

ID	Open question	Owner	Needed before	Affects
OI-01	Which numeric goals for the platform (orders per month, share of orders from the site, conversion rate)?	Client	R1 design	1.2, acceptance
OI-02	Durations and dates for R1, R2, R3	Technical lead	After UI/UX design is approved	2.7
OI-03	Confirm or change the performance, availability and capacity targets in Section 4	Client and technical lead	R1 design	NFR-PERF, NFR-REL
OI-04	Which hosting provider for the backend, database and media, given availability for Syria	Technical lead	R1 start	R-01, DC-04, DC-06
OI-05	Which provider for SMS, OTP and WhatsApp notifications	Technical lead	R1 build	EI-SW-02, EI-SW-06
OI-06	How long must order and payment records be	Client	R1 launch	5.2

ID	Open question	Owner	Needed before	Affects
	kept			
OI-07	Which allergen and labeling rules apply to bakery products	Client	R1 launch	FR-CKE-05
OI-08	Will customers in the EU or UK be served, so GDPR applies	Client	R3	LEG-06
OI-09	How does Sham Cash confirm payments for a merchant: API, callback or manual check	Client	R1 design	FR-PAY-03
OI-10	Can Stripe be used lawfully, and through which legal entity	Client with a lawyer	Before R3 build	FR-PAY-04, R-01
OI-11	Which AI image provider, and what daily budget	Technical lead and client	R3 start	FR-DAI-05
OI-12	Which currency is the base for prices, and who updates the exchange rate each day	Client	R1 design	FR-PAY-05
OI-13	The list of Damascus delivery zones with fees, time slots and slot capacity	Client	R1 build	FR-DLV-02, 04
OI-14	Cut-off time for same-day orders and the lead time for each product type	Client	R1 build	FR-DLV-05
OI-15	The full list of flowers, greens, wraps, ribbons and extras with prices for the designer	Client	R1 build	FR-ADM-03, 5.6

Appendix F: Quality checklist before approval

These checks come from the 29148 characteristics of a good requirement and of a good requirement set.

- ☐ Each requirement is necessary: it traces to a business need in Appendix A.
- ☐ Each requirement is singular, one statement and one idea.
- ☐ No requirement uses vague words (fast, easy, nice, flexible) without a number.
- ☐ Each requirement has an ID, a priority, a release and a verification method.
- ☐ Each requirement is feasible within the budget, schedule and the providers available in Syria.
- ☐ The set is complete: every user class, interface and quality area is covered.
- ☐ The set is consistent: no two requirements contradict each other.
- ☐ Every open issue has an owner and a point by which it must be answered.
- ☐ The client has read the document, answered the open issues and signed the approvals table.